No.	Status	Date
HIS-2013-0402-DS-2-RF-102	Issued	2013-09-09

UT Thickness Data Sheet - 2-RF-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 2-RF-102, located within the ATU-100. The inspection was conducted as part of the 3-year regenerator overhead and reflux survey detailed in HIS-2013-0402.

Kevin Doucette, acting as lead UT technician and SNT-TC-1A UT Level II for Halward Inspection Services Ltd., performed the thickness measurements and data recording for this line.

Technical Specifications

The subject line, 2-RF-102, is a reflux line constructed from A312 TP316 with an outer diameter of 60.3 mm and a nominal wall thickness of 3.91 mm. The design limits for this asset are 517 kPa(g) and 104 °C.

The minimum allowable thickness (t-min) for this line is 2.16 mm. Any measured thickness falling below this value is subject to an NCR.

Inspection Methodology

Thickness measurements were performed in accordance with the requirements of ASTM E797. The field activities were executed under the site NDE procedure QSCP-NDT-001 Rev 2, which incorporates specific requirements for the CUI program, including insulation removal and inspection-port access.

The technical execution of the ultrasonic testing followed the HIS-UT-002 Rev 1 procedure submitted by Halward Inspection Services Ltd. for QSCP acceptance.

Asset Integrity Analysis

The line is categorized as Class 2 for amine, reflux, and treated gas, which carries a thickness measurement interval of 10 years and an external visual interval of 5 years. However, as part of the regenerator overhead circuit, this line is currently on a 3-year thickness interval.

Remaining life and the next inspection date are determined by the larger of the long-term or short-term corrosion rates. The next reading is scheduled for no more than half the remaining life, not exceeding the class maximum.

Thickness readings, measured 2013-09-09

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2010-09-13	6.78	6.49	0.063	0.077	38.1	2016-09-09	
3-RV-101-02	7.62	3.56	2010-09-13	6.59	6.44	0.065	0.075	38.3	2016-09-09	
3-RV-101-03	7.62	3.56	2010-09-13	6.16	5.80	0.098	0.127	17.6	2016-09-09	
3-RV-101-04	7.62	3.56	2010-09-13	6.43	6.13	0.083	0.089	29.0	2016-09-09	
3-RV-101-05	7.62	3.56	2010-09-13	6.63	6.48	0.068	0.065	43.0	2016-09-09	
3-RV-101-06	7.62	3.56	2010-09-13	6.43	6.11	0.085	0.100	25.4	2016-09-09	
3-RV-101-07	7.62	3.56	2010-09-13	6.20	5.73	0.157	0.157	13.8	2016-09-09	
3-RV-101-08	7.62	3.56	2010-09-13	6.71	6.36	0.117	0.117	23.9	2016-09-09	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2010-09-13	4.55	4.49	0.059	0.057	28.6	2016-09-09	
2-RV-102-02	5.54	2.79	2010-09-13	4.60	4.29	0.074	0.064	20.2	2016-09-09	
2-RV-102-03	5.54	2.79	2010-09-13	4.84	4.68	0.046	0.055	34.2	2016-09-09	
2-RV-102-04	5.54	2.79	2010-09-13	4.84	4.81	0.044	0.052	38.9	2016-09-09	
2-RV-102-05	5.54	2.79	2010-09-13	5.04	4.96	0.027	0.027	>50	2016-09-09	
2-RV-102-06	5.54	2.79	2010-09-13	4.82	4.55	0.090	0.090	19.5	2016-09-09	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2006-02-24	5.48	5.39	0.005	0.012	>50	2023-09-09	
3-RF-101-02	5.49	2.41	2006-02-24	5.37	5.34	0.006	0.004	>50	2023-09-09	
3-RF-101-03	5.49	2.41	2006-02-24	5.45	5.45	0.000	0.000	>50	2023-09-09	
3-RF-101-04	5.49	2.41	2006-02-24	5.33	5.33	0.007	0.000	>50	2023-09-09	
3-RF-101-05	5.49	2.41	2006-02-24	5.40	5.26	0.011	0.019	>50	2023-09-09	
3-RF-101-06	5.49	2.41	2006-02-24	5.40	5.25	0.015	0.020	>50	2023-09-09	
3-RF-101-07	5.49	2.41	2006-02-24	5.35	5.43	-0.001	-0.011	>50	2023-09-09	
3-RF-101-09	5.49	2.41	—	—	5.12	—	—	—	2023-09-09	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2006-02-24	3.85	3.71	0.012	0.019	>50	2023-09-09	
2-RF-102-02	3.91	2.16	2006-02-24	3.88	3.65	0.013	0.031	48.8	2023-09-09	
2-RF-102-03	3.91	2.16	2006-02-24	3.80	3.76	0.010	0.005	>50	2023-09-09	
2-RF-102-04	3.91	2.16	2006-02-24	3.81	3.80	0.004	0.001	>50	2023-09-09	
2-RF-102-05	3.91	2.16	—	—	3.69	—	—	—	2023-09-09	
2-RF-102-06	3.91	2.16	—	—	3.71	—	—	—	2023-09-09	
2-RF-102-07	3.91	2.16	2006-02-24	3.85	3.82	0.008	0.004	>50	2023-09-09	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2013-09-09	_____